

# Probability Assignment 3

Name :- Sana Munir Alam

Roll# :- 24K-0513

Class :- BCS-4E

Date: \_\_\_\_\_

Q) Data:-

$$\bar{x} = 1.8$$

$$u = 2.1?$$

$$\sigma = 20 \text{ min} \approx 0.333 \text{ hrs or } \frac{1}{3} \text{ hrs}$$

$$n = 50$$

$$1 - \alpha = 90\%$$

$$\frac{\alpha}{2} = \frac{0.1}{2}$$

$$\frac{\alpha}{2} = 0.05$$

referring to standard Normal Distribution table

$$Z_{\alpha/2} = 1.645$$

Confidence Interval Formula

$$\bar{x} \pm Z_{\alpha/2} \left( \frac{\sigma}{\sqrt{n}} \right)$$

$$1.8 + 1.645 \left( \frac{1/3}{\sqrt{50}} \right) = 1.877546 \approx 1.878$$

$$1.8 - 1.645 \left( \frac{1/3}{\sqrt{50}} \right) = 1.722453 \approx 1.722$$

$\therefore$  90% Confidence interval lies between (1.722, 1.878) hrs

Hence there is evidence that the true mean interruption time for this company is lower than the reported 2.1 hrs, since the reported value does not lie within the 90% confidence interval.

Q2)  $n = 10$

$$\bar{x} = \frac{225 + 240 + 215 + 206 + 211 + 210 + 193 + 250 + 225 + 202}{10}$$

$$\bar{x} = 217.7$$

$$1 - \alpha = 95\%$$

$$\alpha = 1 - 0.95$$

$$\alpha = 0.05$$

$$\frac{\alpha}{2} = 0.025$$

as  $n < 30$  and we aren't given

$\sigma$  we will use  $t$  table

degree of freedom  $\Rightarrow n - 1 = 9$

for  $s \Rightarrow s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$  or  $s = \sqrt{\frac{\sum x^2 - \frac{(\sum x)^2}{n}}{n-1}}$

$$\sum x = 2177$$

$$s^2 = \frac{476685 - \frac{(2177)^2}{10}}{9}$$

$$\sum x^2 = 476685$$

$$\sqrt{s^2} = \sqrt{\frac{27521}{9}}$$

$$s = \cancel{14.46} 17.48$$

for  $\frac{t_\alpha}{2} = t_{0.025} = 2.262$

Confidence Interval Formula

$$\bar{x} \pm t_{\frac{\alpha}{2}} \left( \frac{s}{\sqrt{n}} \right)$$

$$217.7 + 2.262 \left( \frac{17.48}{\sqrt{10}} \right) = 230.20357 \approx 230.20$$

$$217.7 - 2.262 \left( \frac{17.48}{\sqrt{10}} \right) = 205.19643 \approx 205.20$$

$\therefore$  95% Confidence interval (205.20, 230.2)

Hence we are 95% confident that the true mean price of the laptop lies between \$205.20 and \$230.20

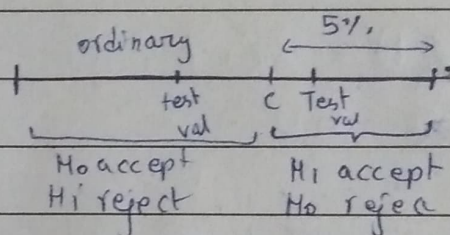
### Q3) Hypothesis Test

$H_1$  (claim) :- Avg salary of entry level software > \$42000

$H_0$  (null hypothesis) :- Mean salary is ~~43260~~ 42000

claim

It's a right tailed test



Data

$$n = 30$$

$$\mu = 42000$$

$$\bar{x} = 43260$$

$$\sigma = 5230$$

} standard deviation

$$\alpha = 0.05$$

} significance level

$$P(X > \text{Test val}) < 5$$

for claim to be accepted

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{43260 - 42000}{\left( \frac{5200}{\sqrt{30}} \right)} = 1.32$$

$$Z_{\alpha} = Z_{0.05} = 1.645$$

$$\text{as } 1.32 < 1.645$$

There is not enough evidence to support the claim that the engineers earn more than 42000 per year.

Q4)  $\alpha = 0.05$  so  $Z_{\alpha} = Z_{0.05} = 1.645$

$$H_0: \mu = \$24 \text{ billion}$$

$$H_1: \mu > \$24 \text{ billion}$$

Note:- Total value was

51 yet  $n$  is 50 so  
I am going with 50

Data

$$\bar{X} = \frac{\text{sum of all value}}{n} = \frac{1476}{50} = 29.52$$

$$s = 28.7$$

$$n = 50$$

Sol  $Z = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} = \frac{29.52 - 24}{28.7 / \sqrt{50}} = 1.36$

It's a right tailed test

$$P = P(Z > 1.36)$$

$$P = 1 - P(Z < 1.36)$$

$$P = 0.0869$$

as  $0.0869 > 0.05$

$\therefore$  There is insufficient evidence at the 5% significance level to support the claim that the average annual revenue of tech companies exceeds \$24 billion.

Q25)

$$H_0: \mu = 5.8$$

$$H_1: \mu \neq 5.8$$

{ Two tailed test

$$\alpha = 0.05 \rightarrow \text{as it's two tailed so } 0.025$$

$$n = 20$$

$$\bar{x} = \frac{\sum x}{n} \Rightarrow \sum x = 77$$

$$\sum x^2 = 417$$

so

$$\bar{x} = \frac{\sum x}{n} = \frac{77}{20} = 3.85$$

$$S^2 = \frac{\sum x^2 - \frac{(\sum x)^2}{n}}{(n-1)} = \frac{417 - \frac{77^2}{20}}{19} = 6.344$$

$$\sqrt{S^2} = \sqrt{6.344}$$

$$S = 2.51$$

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{3.85 - 5.8}{2.51/\sqrt{20}} = -3.474$$

$$\text{degree of freedom} \Rightarrow v = n - 1 = 19$$

$$\text{so } t_{\frac{\alpha}{2}} = t_{0.025} = 2.093$$

$$\text{so right side} = +2.093$$

$$\text{left side} = -2.093$$

$$\text{rejection region } t < -2.093 \text{ or } t > 2.093$$

$$\text{as } |t| = 3.71 > 2.093 \therefore \text{lies in rejection region}$$

Hence there is sufficient evidence at the 5% significance level to conclude that the average app usage is not equal to 5.8 times per

day. In fact, it's significantly lower based on the sample.

Q6) Data Algo 1

$$n = 12$$

$$\sigma = 4$$

$$\bar{x} = 85$$

Data Algo 2

$$m = 10$$

$$\sigma = 5$$

$$\bar{x} = 81$$

assumption:- equal variance

$$\alpha = 0.05$$

$$H_0 : \mu_1 - \mu_2 \leq 2$$

$$H_1 : \mu_1 - \mu_2 > 2$$

as per assumption:-  $\sigma_x^2 = \sigma_y^2 = \sigma^2$   
calculate pooled sample variance

$$S_p^2 = \frac{(n-1)S_x^2 + (m-1)S_y^2}{n+m-2}$$

$$S_p^2 = \frac{(12-1)(4)^2 + (10-1)(5)^2}{12+10-2}$$

$$\sqrt{S_p^2} = \sqrt{20.05} \Rightarrow S_p = 4.477$$

~~Confidence Interval~~

~~$$\bar{x} - \bar{y} \pm t_{\alpha/2} S_p \sqrt{\frac{1}{n} + \frac{1}{m}}$$~~

$$t = \frac{(\bar{x} - \bar{y}) - 2}{S_p \sqrt{\frac{1}{n} + \frac{1}{m}}} = \frac{(85 - 81) - 2}{4.477 \sqrt{\frac{1}{12} + \frac{1}{10}}} = \frac{2}{1.0433}$$

$$\text{degree of freedom} = n + m - 2 = 12 + 10 - 2 = 20$$

$$t_{0.05, 20} = 1.725$$

$$\text{as } 1.04 < 1.725$$

There is not enough statistical evidence at the 0.05 level to conclude that algorithm 1's performance degradation exceeds Algorithm's 2's by more than 2 units.